

Precedence — full levels inventory

Source inventory: 3 October 2026 (ICT). Cross-checked against `engine.js`, `app.html`, `levels.js`, all `levels-*.js` extension files, `README.md`, `docs/STRUCTURE.md`, and `docs/PRECEDENCE_ERA_UNLOCK_MAP.md`.*

This document separates the **Precedence core** (the order-of-operations racer) from the older/larger **Algebra Licenses** ladder that still ships in the same engine and is hidden from the MVP home.

1. Recommended nomenclature

Use **Eras** for the ten large Precedence garage cards: an Era carries a math title, a rule progression, a world, and an earned visual travel reward. Use **Levels** for the individual problem bands inside the order-of-operations core; a level produces the clean runs/stamps that help clear an Era. Use **Licenses** for the separate Algebra Licenses wing, where each license contains its own named level ladder (for example, Powers 2.1–2.12 and Quadratics 9.1–9.15). Within a playable level, call a single generated problem a **run** or **challenge**, and call each tap/collapse a **step**. This keeps transport as art and progression language, while math remains the visible title system.

2. Precedence core — order-of-ops racer

Status key: *Playable* means wired in the current garage. *Stub* means the card exists but has no Era-specific generator yet. “Engine mapping” is the current `PrecEngine.LEVELS` number; a dash means the future Era is not mapped yet.

#	Math title	Playable?	Engine mapping	Operations / rule focus	Example	Mathera skill IDs	Visual vehicle (development note only)
1	Sums	Yes	L1 — Sums	+ and −; brackets are part of the rule set	$9 - 4 + 3$	II.8.01, III.4.03; bracket rule also III.1.08	On foot; Mesa Verde cliffs
2	Products	Yes	L3 — Products	Adds ×; multiplication before addition/subtraction	$2 + 3 \times 4$	II.8.01, II.8.03, III.4.03	Royal litter / Cleopatra likeness; Alexandria. Visual only; no transport label copy
3	Differences	Yes	L2 — Differences	+ − mix with bracket-order traps; brackets change order	$(2 + 3) \times 4$	II.8.01, III.1.08, III.4.03	Horseback; desert West
4	Quotients	Yes	L4 — Quotients	Adds ÷; × and ÷ resolve left-to-right	$12 \div 3 \times 2$	II.8.01, II.8.03, III.4.03	Horse-drawn carriage; gaslit Boston
5	Mixed operations	No — stub card	Future mix / no current mapping	All four operations in one road test; precedence and left-to-right review	$18 - 6 \div 2 + 1$	Current family: II.8.01, II.8.03, III.4.03; add/confirm any mixed-review IDs during content pass	Early car / Model T; Detroit
6	Powers	No — stub card	L5 — Powers exists, but POWERS = false globally	Adds powers; powers outrank multiplication/division	$2 + 3^2$	Current engine metadata: II.8.01, II.8.03, III.4.03; bracket rule III.1.08 where relevant	Streamliner; Art Deco New York
7	Long chains	No — stub card	L6 — Long chains exists, but is not mapped to the garage	Longer expressions combining brackets and powers	$(1 + 2)^2 \times 2$	Current engine metadata: II.8.01, II.8.03, III.4.03; bracket rule III.1.08	1950s cruiser; Route 66
8	Nested	No — stub card	Future nested generator	Nested brackets; deeper tree/order traps	$2 \times (3 + (4 - 1))$	Current family: II.8.01, II.8.03, III.1.08, III.4.03	Wedge sports car / Miami Burnout; reserved for 1.0
9	Negatives	No — stub card	No current core mapping	Negative numbers inside precedence expressions	$-3 + 4 \times (-2)$	Extend the current order-of-operations family with the integer/negative skill IDs during content pass	Electric vehicle; San Francisco fog
10	Fraction bars	No — stub card / elite	No current core mapping; sister Fractions app exists	Fraction bars plus the full precedence rule set	Compound fraction-bar expression	Extend with fraction-operation IDs; hand-off to Fractions app is an option	Robotaxi; 2040 neon city

Current core wiring

The live `app.html` maps the first four garage cards deliberately as **Sums** → **engine L1**, **Products** → **L3**, **Differences** → **L2**, **Quotients** → **L4**. Products comes before Differences in the garage because the current fantasy ladder introduces × before the dedicated + − bracket-drill band; this is a progression choice, not a claim that the original engine numbering changed. Eras 5–10

are present as locked cards only. The road currently has theme/color packs, not the final foot, litter, horse, carriage, or later vehicle sprites.

The engine does support `^`, and engine levels 5–6 are named **Powers** and **Long chains**, but `const POWERS = false` currently prevents powers from appearing in generated core expressions. The future content pass should turn that on deliberately per Era rather than silently treating the six engine levels as ten finished Eras.

3. Original order-of-operations engine levels 1–6

These are the names and generator definitions in `engine.js`; they are the crosswalk for the current Precedence core.

Engine level	Original display name	Operators / generator intent	Current state
1	Sums	<code>+</code> ; bracket-aware expression trees	Wired to Era 1; playable
2	Differences	<code>+</code> , <code>-</code> ; mixed plus/minus and bracket traps	Wired to Era 3; playable
3	Products	<code>+</code> , <code>-</code> , <code>*</code> ; multiplication precedence	Wired to Era 2; playable
4	Quotients	<code>+</code> , <code>-</code> , <code>*</code> , <code>/</code> ; exact whole-number division and <code>x÷</code> order	Wired to Era 4; playable
5	Powers	Same four base operators plus <code>^</code> when enabled	Exists in engine; not garage-wired; powers globally off
6	Long chains	Longer trees (extra: <code>2</code>) plus powers when enabled	Exists in engine; not garage-wired

Every engine level currently carries the core metadata IDs `II.8.01` and `III.4.03`; levels 3–6 also carry `II.8.03`. The UI rule board adds `III.1.08` for the brackets-first rule. The exact future ID set for negatives and fraction bars should be confirmed against the Mathera skill tree when those generators are written.

4. Algebra Licenses wing — hidden from the Precedence MVP home

This is the compact crosswalk of `LIC` in `levels.js`. The number before the dot is the license; the number after the dot is the level's position in that license. The IDs are stable save keys, so they should not be renumbered casually. “Mixed review” is a review level, not a new topic. Levels marked **comeback** wait for a later license; they remain part of the inventory even when not initially available.

License 2 — Powers

Level ID	Display name
2.1 · 2.mul	Same base, multiply
2.2 · 2.div	Same base, divide
2.3 · 2.pow	Power of a power
2.4 · 2.neg	Zero and negative exponents
2.5 · 2.frac	Fractional exponents
2.6 · 2.laws	All the laws at once
2.7 · 2.negbase	Negatives and fractions as bases
2.8 · 2.negfrac	Negative exponents in fractions
2.9 · 2.sci	Scientific notation
2.10 · 2.mix	Mixed review
2.11 · 2.rad	Radicals and fractional exponents — comeback after 11
2.12 · 2.calc	Rewrite for the power rule — comeback after 13

License 3 — Simplify

Level ID	Display name
3.1 · 3.like	Like terms
3.2 · 3.dist	Distribute
3.3 · 3.minus	Minus before a bracket
3.4 · 3.nest	Nested brackets
3.5 · 3.pow	Like terms with powers
3.6 · 3.fraccoef	Fractions as coefficients
3.7 · 3.twolet	Two letters
3.8 · 3.dec	Decimals
3.9 · 3.mix	Mixed review
3.10 · 3.rat	Cancel factors in fractions — comeback after 10
3.11 · 3.radlog	Radicals and log laws — comeback after 11
3.12 · 3.calc	Tidy a derivative — comeback after 13

License 4 — Expand

Level ID	Display name
4.1 · 4.mono	Monomial times polynomial
4.2 · 4.foil	Two binomials
4.3 · 4.spec	Special products
4.4 · 4.tri	Binomial times trinomial
4.5 · 4.three	Three factors
4.6 · 4.twolet	Two letters
4.7 · 4.expsimp	Expand and simplify
4.8 · 4.cube	Cube a binomial
4.9 · 4.mix	Mixed review
4.10 · 4.conj	Conjugates — comeback after 11
4.11 · 4.dq	Difference quotients — comeback after 12

License 5 — Factor

Level ID	Display name
5.1 · 5.gcf	Common factor
5.2 · 5.gcfpow	Common factor with powers
5.3 · 5.group	Grouping
5.4 · 5.tri1	$x^2 + bx + c$
5.5 · 5.tria	$ax^2 + bx + c$
5.6 · 5.spec	Special patterns
5.7 · 5.gcfthen	Common factor first, then factor
5.8 · 5.twolet	Two letters
5.9 · 5.cubes	Cubes
5.10 · 5.qtype	Quadratic type
5.11 · 5.twice	Factor twice
5.12 · 5.mix	Factor completely
5.13 · 5.divthm	Long division and the factor theorem — comeback after 10
5.14 · 5.fracexp	Negative and fractional exponents — comeback after 11
5.15 · 5.calc	Tidy a derivative — comeback after 13

License 6 — Isolate

Level ID	Display name
6.1 · 6.one	One step
6.2 · 6.two	Two steps
6.3 · 6.brack	Brackets first
6.4 · 6.both	x on both sides
6.5 · 6.frac	Fractions and decimals
6.6 · 6.prop	Proportions
6.7 · 6.none	No solution or every number
6.8 · 6.abs	Absolute value
6.9 · 6.form	Formulas
6.10 · 6.twice	The letter shows up twice
6.11 · 6.bottom	x on the bottom
6.12 · 6.mix	Mixed review
6.13 · 6.ratq	Rational equations that turn quadratic — comeback after 10
6.14 · 6.rad	Radical equations and false roots — comeback after 11
6.15 · 6.explog	Exponential and log equations — comeback after 11
6.16 · 6.calc	Where the slope is zero — comeback after 13

License 7 — Inequalities

Level ID	Display name
7.1 · 7.one	One step
7.2 · 7.flip	Negative flips
7.3 · 7.two	Two steps
7.4 · 7.comp	Compound
7.5 · 7.abs	Absolute value
7.6 · 7.nl	Number line
7.7 · 7.int	Interval notation
7.8 · 7.xy	Two variables
7.9 · 7.both	Where both are true
7.10 · 7.frac	Fractions in inequalities
7.11 · 7.mix	Mixed review
7.12 · 7.rat	Rational inequalities — comeback after 10
7.13 · 7.calc	Where a function rises — comeback after 13

License 8 — Systems

Level ID	Display name
8.1 · 8.sub	Substitution
8.2 · 8.add	Add to eliminate
8.3 · 8.scale	Scale, then eliminate
8.4 · 8.none	None or infinitely many
8.5 · 8.three	Three unknowns
8.6 · 8.line	Pick the line
8.7 · 8.graph	Graph to solve
8.8 · 8.slopes	Slopes tell the story
8.9 · 8.frac	Fractions and decimals
8.10 · 8.mix	Mixed review
8.11 · 8.parab	Line meets parabola — comeback after 9
8.12 · 8.matrix	Matrices and row reduction — comeback after 16

License 9 — Quadratics

Level ID	Display name
9.1 · 9.sqrt	Square roots
9.2 · 9.fact	Factoring
9.3 · 9.rearr	Move everything to one side first
9.4 · 9.cts	Complete the square
9.5 · 9.formula	The formula
9.6 · 9.disc	How many solutions?
9.7 · 9.method	Pick your method
9.8 · 9.vertex	Vertex form
9.9 · 9.qtype	Quadratic type
9.10 · 9.build	Build it from its roots
9.11 · 9.read	Read the parabola
9.12 · 9.ineq	Quadratic inequalities
9.13 · 9.linepar	Line meets parabola
9.14 · 9.mix	Mixed review
9.15 · 9.disguise	Quadratics in disguise — comeback after 11
9.16 · 9.calc	Highest and lowest points — comeback after 13

License 10 — Rationals

Level ID	Display name
10.1 · 10.cancel	Cancel common factors
10.2 · 10.muldiv	Multiply and divide
10.3 · 10.same	Same bottom
10.4 · 10.lcd	Different bottoms
10.5 · 10.complex	Fraction over fraction
10.6 · 10.eq	Rational equations
10.7 · 10.longdiv	Long division
10.8 · 10.remthm	The remainder theorem
10.9 · 10.graph	Read the graph
10.10 · 10.partial	Split into partial fractions
10.11 · 10.mix	Mixed review

License 11 — Radicals & logs

Level ID	Display name
11.1 · 11.simp	Simplify a root
11.2 · 11.muldiv	Multiply and divide roots
11.3 · 11.add	Add like roots
11.4 · 11.ration	Rationalize the bottom
11.5 · 11.fracexp	Roots and fractional exponents
11.6 · 11.tolog	Power to log and back
11.7 · 11.eval	Work out a log
11.8 · 11.expand	Expand with the log laws
11.9 · 11.condense	Condense into one log
11.10 · 11.base	Change of base
11.11 · 11.graph	Read the graph
11.12 · 11.mix	Mixed review

Sister app: LIC[0] — Fractions / Bakery Lane

LIC[0] is intentionally not an Algebra License. It is the separate Fractions app ladder (storage key `fractions-v1`, level codes F1–F17) and remains the intended sister app for the late Fraction bars Era.

ID	Display name
f.1 · f.name	Name the shaded part
f.2 · f.line	Fractions on a number line
f.3 · f.equiv	Equivalent fractions
f.4 · f.simp	Simplify
f.5 · f.cmp	Compare and order
f.6 · f.mixed	Mixed numbers
f.7 · f.same	Add and subtract, same bottom
f.8 · f.diff	Add and subtract, different bottoms
f.9 · f.mixop	Mixed numbers with regrouping
f.10 · f.mul	Multiply, cancel first
f.11 · f.div	Divide
f.12 · f.of	A fraction of an amount
f.13 · f.fdp	Fractions, decimals, percents
f.14 · f.order	Order of operations
f.15 · f.letter	The same moves with a letter
f.16 · f.cancel	Cancel factors, never terms
f.17 · f.mix	Mixed review

5. Gaps and mismatches to resolve in the content pass

- Garage order differs from engine order.** The current MVP intentionally shows Products before Differences (Era 2 uses engine L3; Era 3 uses L2). Keep the math titles, but document the progression choice in code so nobody “fixes” it back accidentally.
- Powers are implemented but switched off.** `engine.js` supports powers and names L5/L6, while `POWERS = false` means generated core play is still arithmetic-only. Era 6 should get an explicit powers-on test before being called playable.

3. **Ten Eras are not ten finished content bands yet.** Eras 5–10 are garage stubs; only 1–4 are currently wired to playable generators. Era 8 / Nested is the Miami Burnout 1.0 milestone, not a current feature.
4. **Current visual transport is placeholder art.** `road.js` has world/color packs, but foot, Cleopatra's litter, horse, and carriage sprites still need to replace the car silhouettes. Cleopatra can remain a visual likeness with no explanatory transport copy.
5. **Skill coverage is thin in the core metadata.** Most engine levels point to `II.8.01`, `II.8.03`, and `III.4.03`; the UI's brackets-first rule also names `III.1.08`. This is enough for the shell, not a complete map for mixed operations, negatives, powers, or fraction bars.
6. **Era 3 is a content mismatch.** Its current engine mapping is L2, but the live L2 generator is primarily `+` `-` and does not yet constitute a dedicated bracket-trap band matching the design example. A focused generator/content pass is still needed.
7. **Era 4 has the right operator set but not a separate quotient curriculum.** L4 supplies exact division and `×÷` precedence; verify the question mix and avoid presenting every quotient as merely another generic tree.
8. **Fraction bars have an architectural boundary.** `LIC[0]` is a strong sister Fractions app, but no in-core Era 10 generator is present. Decide whether Era 10 is native, a showcase hand-off, or both.
9. **Algebra Licenses are much broader than Precedence.** The 122-level Algebra Licenses inventory and its comeback rules should stay hidden from the Precedence garage until the core progression is stable.
10. **Naming should remain stable in saves.** User-facing names can be refined, but stable IDs such as `5.gcf` and `11.fracexp` are save keys and migration inputs.

Quick implementation checklist

- Keep the home cards titled **Sums, Products, Differences, Quotients, Mixed operations, Powers, Long chains, Nested, Negatives, Fraction bars**.
- Keep transport out of the math title and out of the explanatory UI copy where Julius requested visual-only treatment.
- Treat Eras 1–4 as the first playable structure build; do content refinement against the original demo after the shell is accepted.
- Turn on and test powers only when Era 6 content is ready; do not infer readiness from the existence of L5/L6 names.
- Reserve **Nested / Miami Burnout** for the 1.0 milestone.